



Automatic Mood-Conditioned Symbolic Music Generation

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Project Overview

We study the problem of generating symbolic music that matches a specified emotional mood. Using labeled MIDI data, we build models that generate pitch sequences conditioned on emotions such as happy, sad, calm, and excited. We first implement a mood-conditioned LSTM as a baseline to capture local temporal structure. We then extend this approach with Transformer-based architectures to better model long-range musical dependencies. Our models output playable MIDI files that can assist musicians in emotion-driven composition workflows.

Dataset & Features

We use the XMIDI dataset, containing 108,023 MIDI files, which use the following filename convention to denote the labeled emotion and genre:

XMIDI_<emotion>_<genre>_<ID_len_8>

Example: XMIDI_angry_classical_OHP7PK58

The dataset contains about 5,278 hours of audio total with an average duration of 176 seconds.

LSTM Implementation

Each MIDI file is parsed into fixed-length token note sequence using pitch values only, with emotions encoded as multi-hot vectors derived from the filename.

Transformer Implementation

Each MIDI file is converted into a rich symbolic event sequence using a REMI tokenizer, which encodes musical structure as discrete tokens for note pitches, note durations, time shifts, velocities, instrument programs, chords, and tempo changes. Mood and genre labels are derived from the filename and injected as a special <mood_genre> token prepended to each sequence. The model is explicitly conditioned on the target emotion from the first step of generation. The resulting mood-conditioned REMI event sequences are then mapped to integer token IDs.

Model	Train/Val Split	Total # in Train/Val
LSTM	90 / 10	1,800 / 200
Transformer	91 / 9	50,000 / 5,000

Table 1: Summary of Datasets

Models

LSTM Architecture

We implement a mood-conditioned LSTM that concatenates token embeddings with emotion embeddings to predict the next musical event. Training is performed using teacher forcing with gradient clipping, and model selection is based on minimum validation loss. Padding tokens are excluded during loss computation.

Epochs	Batch Size	Optimizer	Learning Rate	Hidden Size	Layers	Dropout	Loss
20	32	AdamW	1×10^{-3}	512	3	0.3	Categorical Cross-Entropy

Table 2: Hyperparameters Used For LSTM

Transformer Architecture

We then implement a Transformer decoder with self-attention and learned mood memory tokens to model long-range musical structure. We trained using the same teacher forcing strategy used in the LSTM baseline. At each time step, the model predicts the next MIDI token given all previously observed tokens and four mood-conditioning slots. As in the baseline, padding tokens are excluded during loss computation.

Epochs	Batch Size	Optimizer	Learning Rate	Hidden Size	Attention Heads	Layers	Dropout	Loss
10	32	AdamW	1×10^{-3}	256	4	3	0.1	Cross-Entropy

Table 3: Hyperparameters Used For the Transformer-Based Model

Results

LSTM Results

The LSTM exhibits stable convergence with closely aligned training and validation loss curves, indicating limited overfitting. Conditioning on emotion enables the model to generate stylistically consistent melodic sequences.

Epoch	Train Loss	Val Loss	Val Macro-AUROC
5	0.98	1.07	0.79
10	0.55	0.63	0.82
15	0.41	0.44	0.84
20	0.37	0.40	0.845

Table 4: LSTM Results Table

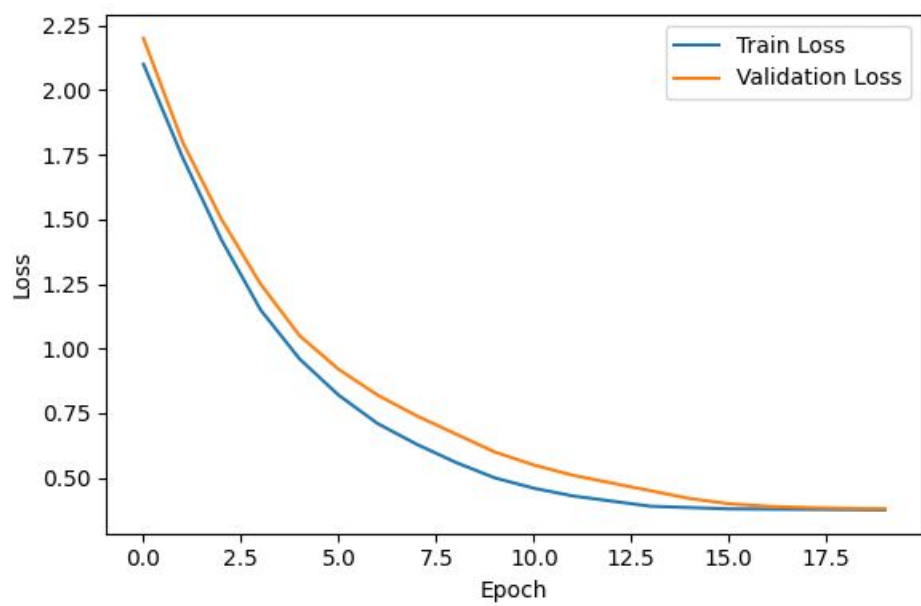


Figure 1: LSTM Training Loss Curve

Transformer Results

The Transformer converges very rapidly, with validation loss dropping from ~2.0 to near 0.0 within the first 10 epochs. However, models that reached very low validation losses (below 0.4–0.8) collapsed to a degenerate solution that repeatedly predicted only a small subset of pitch tokens, yielding MIDI files with almost no audible music. To avoid this mode collapse, we employed early stopping based on validation loss and qualitative listening tests, selecting checkpoints that balanced numerical performance with musically diverse, non-trivial generations.

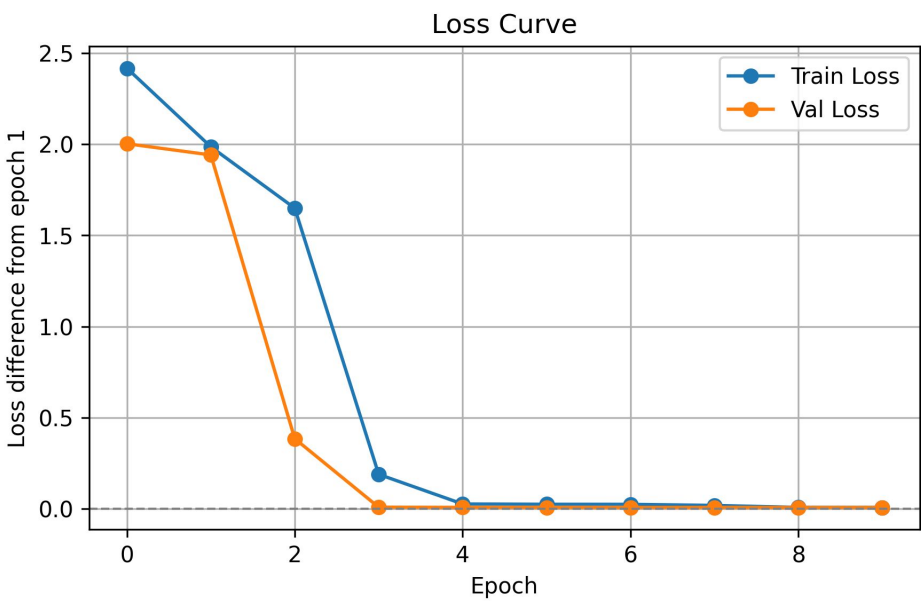


Figure 2: Transformer Training Loss Curve

Discussion

Ultimately, both models were able to generate audio sequences through prediction following training. However, qualitative evaluation revealed that this audio was often sparse and consisting of very few instrumental sounds.

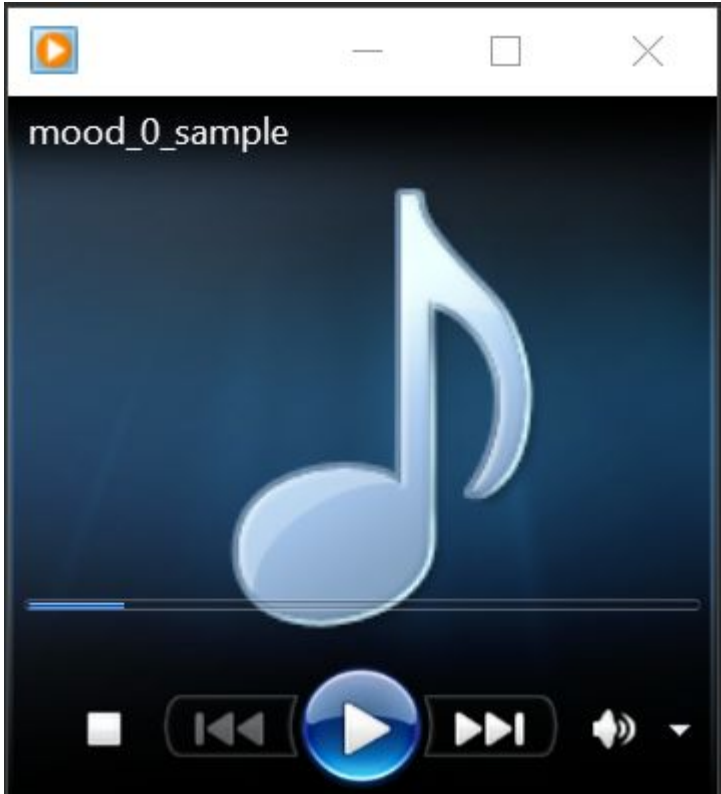


Image 1: Playable MIDI Output

This is likely because the LSTM baseline effectively captured short-term melodic structure but struggled with long-range musical coherence, such as repeating motifs and phrase-level organization. Conditioning on emotion improved stylistic consistency, but emotional expression remained subtle in some generated samples.

The transformer model significantly improved global structure and rhythmic consistency, but we believe it also struggled to capture long-term musical structure due to its shorter, often uni-instrumental output. Training revealed practical challenges in converting token predictions into interpretable MIDI files even with careful tokenization and post-processing.

Despite these challenges, our models show promise that mood-based music generation is feasible, and with more time, we believe we could produce output that is more useful to those in the music industry.

Future Work

With more time and resources, targeting audio cohesion via richer tokenizations that encode rhythm, duration, and polyphonic structure more explicitly would be our priority. We would also aim to model each individual instrument's voice separately. tracks separately to improve musical coherence. Scaling training to larger datasets and incorporating human evaluation would further strengthen emotional expressiveness.

References

[1] @article{xmusic2025,
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